

Manvi Vijay Gawande

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MS Data Science candidate at Indiana University Bloomington (May 2026) with hands-on experience building multi-agent AI systems, RAG pipelines, and NLP models at scale. Seeking full-time roles in Machine Learning Engineering, Applied AI, or Data Science.

EDUCATION

Indiana University, Bloomington	May 2026
Master of Science in Data Science	GPA: 3.856/4.0
Relevant Coursework: Applied Machine Learning, Intro to LLMs, Social Media Mining, Network Science, Data Visualization	
Savitribai Phule Pune University	May 2024
Bachelor of Engineering in Computer Engineering, Honors in Machine Learning and Artificial Intelligence	GPA: 3.81/4.0
Relevant Coursework: Machine Learning, Data Science & Big Data Analytics, Cloud Computing, Database Management Systems. Awarded Excellence in Academics for top score among 160 students.	

WORK EXPERIENCE

Applied ML Research Analyst at Indiana University Bloomington	January 2026-Present
Engineered end-to-end web scraping pipelines to collect and preprocess ~75,000 multimodal records on ivory artifacts and trade, enabling structured analysis of large-scale, heterogeneous historical and transactional datasets.	
Developed recommendation systems and predictive models to identify patterns across complex ivory datasets; implemented a QWEN-powered RAG pipeline to streamline qualitative research workflows and accelerate domain-specific insight generation.	
Research Assistant at Soda Lab	September 2025-Present
Analyzed 15 years of Reddit discourse (2008–2023) across 500K+ posts using LLMs and embedding models to detect belief formation patterns and behavioral shifts in online communities.	
Built an NLP pipeline leveraging transformer-based embeddings to identify and predict mental-health-related signals from large-scale social media data, surfacing statistically significant linguistic markers.	
Data Analyst Intern at S.K. Technovations	November 2022-February 2023
Performed exploratory data analysis on a dataset of over 20,000 rows, utilizing SQL for data extraction, transformation, and loading (ETL) to identify and streamline trends in customer behavior.	
Automated data quality workflows, cutting processing time by 30% and saving 8 hours weekly.	
Data Science Intern at Let's Grow More	August 2022-October 2022
Built a sales predictive model that increased accuracy by 13%, achieving an overall model accuracy of 91.75%	
Created 15+ Tableau dashboards, delivering key insights from large-scale sales data (500K+ rows) that influenced quarterly strategy and improved revenue tracking. Used advanced Excel functions for preliminary data cleaning and trend analysis before dashboard design.	

PROJECTS

FinGuard AI: Adaptive, Explainable Fraud Detection System
Technologies used: Python, LLaMA 3.1, Thompson Sampling, RAG, Async Multi-Agent Architecture
Built FinGuard AI, a multi-agent fraud detection framework that performs continuous online A/B testing across 4 detection-explanation strategy pairs, using Thompson Sampling to adapt in real time and converge to dominant strategies within ~1-2k transactions without offline retraining.
Achieved 8-15% improvement in precision at fixed recall and ~20% reduction in false positives over LLM-only baselines on highly imbalanced datasets (Kaggle Credit Card Fraud, PaySim), while generating 100% auditable, regulation-grounded explanations via RAG over FinCEN and U.S. AML guidance.
Multi-Model Energy Consumption Forecasting and Benchmarking
Technologies used: Python, PyTorch, Scikit-learn, XGBoost, LightGBM, Optuna, Statsmodels, Streamlit, Pandas, Matplotlib, Seaborn
Engineered 50+ temporal and weather-based features for 19k+ records, driving an 18% MAPE improvement over statistical baselines.
Optimized an ML stack (XGBoost, LightGBM, LSTM) via Optuna, slashing the error rate from 153% to 22.3% by mitigating overfitting.
Developed an end-to-end pipeline and Streamlit dashboard to visualize real-time forecasts and benchmarks for a PyTorch-based LSTM.
E-Commerce Customer Segmentation and RFM Analysis
Technologies used: SQL (SQLite), Python (Pandas, Numpy, Plotly), A/B Testing, Tableau, Streamlit
Architected a SQL-based data pipeline for 100k+ records, using complex relational joins and feature engineering to facilitate behavioral analysis.
Engineered an RFM model to segment 96k+ customers into five personas, establishing a framework for A/B testing personalized retention and growth campaigns.
Deployed a full-stack BI suite using Tableau and Streamlit to visualize geographic revenue hubs and marketplace performance metrics for stakeholders.
ArchEvo: Dataset-Aware Neural Architecture Search
Technologies used: Python, PyTorch, Streamlit, SQLite, NetworkX, torchvision, Matplotlib
Architected a Neural Architecture Search (NAS) framework utilizing DARTS and Evolutionary algorithms to optimize 14-edge cell DAGs, achieving 86% accuracy on EuroSAT and 71% on CIFAR-10 within a 1M-parameter budget.
Developed a pressure-aware system with 4 constraint modes (e.g., memory, latency) to enable hardware-targeted architecture discovery using a tunable penalty weight.
Built a Streamlit dashboard with live search tracking, SQLite-based experiment logging, and a surrogate recommender to predict optimal genotypes from dataset profiles.

TECHNICAL SKILLS

Languages: Python, SQL, R, NoSQL
ML & AI: Machine Learning, Deep Learning, NLP, Retrieval-Augmented Generation (RAG), Multi-Agent Systems, A/B Testing, Statistical Modeling
Frameworks & Libraries: HuggingFace, LLaMA, QWEN, Sentence-Transformers, Milvus, FastAPI, Apache Spark
Tools & Platforms: Tableau, Power BI, AWS, Docker, MS Excel

EXTRACURRICULAR ACTIVITIES

Machine Learning Head, Google Developer Student Club, XPCCOER	August 2022-September 2023
Organized workshops or training sessions for 200 students on machine learning for club members, fostering skill development and interest in the field.	
3rd Place, Luddy Hackathon – Indiana University Bloomington	April 2025
Awarded 3rd place for developing Bibble, an AI-powered meeting companion that transforms multilingual video meetings into speaker-attributed transcripts, summaries, and emotional insights.	